

Topic 9 – Ecosystems and material cycles

Students should:		Maths skills
9.1	Describe the different levels of organisation from individual organisms, populations, communities, to the whole ecosystem	
9.2	Explain how communities can be affected by abiotic and biotic factors, including: a temperature, light, water, pollutants b competition, predation	4a, 4c
9.3	Describe the importance of interdependence in a community	
9.4	Describe how the survival of some organisms is dependent on other species, including parasitism and mutualism	
9.5	<i>Core Practical: Investigate the relationship between organisms and their environment using field-work techniques, including quadrats and belt transects</i>	1c, 1d, 2b, 2c, 2d, 2f, 2g, 4a, 4c
9.6	Explain how to determine the number of organisms in a given area using raw data from field-work techniques, including quadrats and belt transects	1c, 1d 2b, 2c, 2d, 2g 4a, 4c
9.7B	Explain how some energy is transferred to less useful forms at each trophic level and that this affects the number of organisms at each trophic level, limits the length of a food chain and determines the shape of a pyramid of biomass in an ecosystem	
9.8B	Calculate the efficiency of energy transfers between trophic levels and percentage calculations of biomass	1a, 1b, 1c 2c 4a
9.9	Explain the positive and negative human interactions within ecosystems and their impacts on biodiversity, including: a fish farming b introduction of non-indigenous species c eutrophication	2c, 2g 4a, 4c
9.10	Explain the benefits of maintaining local and global biodiversity, including the conservation of animal species and the impact of reforestation	

Students should:	Maths skills
9.11B Describe the biological factors affecting levels of food security, including: - a increasing human population - b increasing animal farming and the increased meat and fish consumption - c the impact of new pests and pathogens - d environmental change caused by human activity - e sustainability issues, e.g. use of land for biofuel production and the cost of agricultural inputs	2c 4a, 4c
9.12 Describe how different materials cycle through the abiotic and biotic components of an ecosystem	
9.13 Explain the importance of the carbon cycle, including the processes involved and the role of microorganisms as decomposers	
9.14 Explain the importance of the water cycle, including the processes involved and the production of potable water in areas of drought including desalination	
9.15 Explain how nitrates are made available for plant uptake, including the use of fertilisers, crop rotation and the role of bacteria in the nitrogen cycle	
9.16B Evaluate the use of indicator species as evidence to assess the level of pollution, including: a polluted water – bloodworm, sludgeworm b clean water – freshwater shrimps, stonefly c air quality – different species of lichen, blackspot fungus on roses	2c, 2g 4a, 4c
9.17B Explain the effects of temperature, water content and oxygen availability on the rate of decomposition in food preservation	
9.18B Explain the effects of temperature, water content and oxygen availability on the rate of decomposition in composting	2c 4a, 4c
9.19B Calculate rate changes in the decay of biological material	1c 2c, 2f 4a, 4c

Use of mathematics

- Calculate surface area : volume ratios (1c).
- Plot, draw and interpret appropriate graphs (4a, 4b, 4c and 4d).
- Understand and use percentiles and calculate percentage gain and loss of mass (1c).
- Translate information between numerical and graphical forms (4a).
- Construct and interpret frequency tables and diagrams, bar charts and histograms (2c).
- Understand the principles of sampling as applied to scientific data (2d).
- Use a scatter diagram to identify a correlation between two variables (2g).
- Calculate the percentage of mass (1c).
- Use fractions and percentages (1c).
- Calculate arithmetic means (2b).
- Calculate the rate changes in the decay of biological material (1c).
- Extract and interpret information from charts, graphs and tables (2c, 4a).

Suggested practicals

- Investigate tropic responses.
- Investigate how indicator species can be used to assess levels of pollution in water or the atmosphere.
- Investigate animal behaviour using choice chambers.